

Braille Peg Cell

Field guide for ChipuRobo school visits · tactile object & 3D printing

Model: Braille Peg Cell by **3D Printy** — Printables model 146102

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Cost: free STL download · printables.com/model/146102-braille-peg-cell

What it is

A single braille cell — six dots — where each dot is a **button that pushes through from either side**. Press a dot from below and it stands proud. Press it from above and it drops flat.



That is the whole idea, and it is a good one. A learner can **set** a letter with their own fingers instead of only reading one that has been printed for them — then feel it, clear it, and set the next.

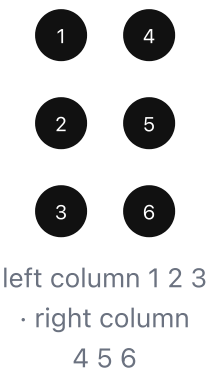
Why it matters for us

Everything else we hand a learner is fixed. A printed skull is always a skull. A braille sheet always says the same thing. This is the first object a learner can **change**.

- A blind learner can **write** as well as read — the loop closes.
- A sighted teacher who does not know braille can learn alongside the class. In a mainstream school that is usually the real blocker.
- Cheap enough to print **one per learner** rather than one per classroom. Sharing a tactile object means most of the class waits.

The cell, and how the dots are numbered

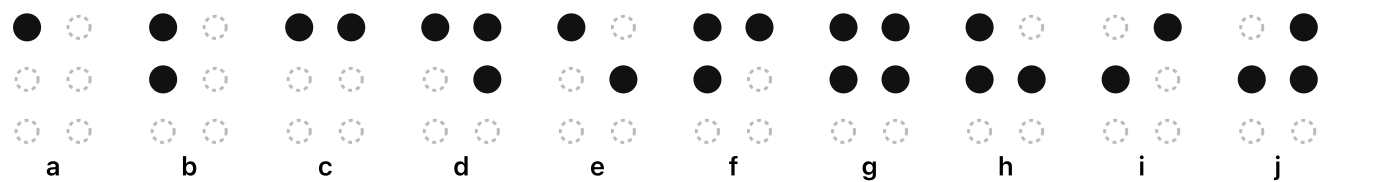
Six dots, two columns of three. The **numbering** is what everything else is built on. Teach it before any letters — drill it until it is automatic, and every letter afterwards is just a list of numbers.



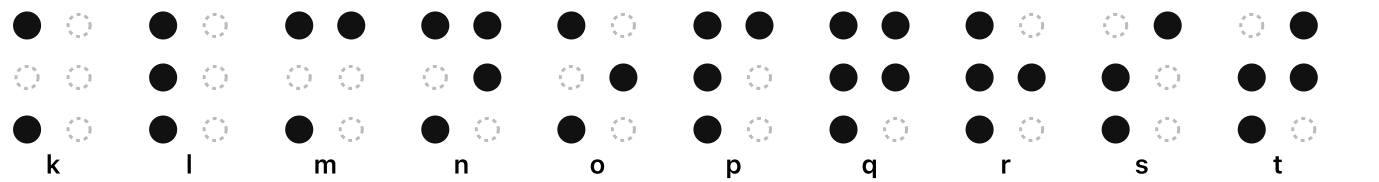
The first ten letters use only the top four dots (1, 2, 4, 5). Learn these ten and two rules, and you have the whole alphabet — far easier than memorising twenty-six unrelated shapes.

k–t	the same shapes as a–j, plus dot 3
u, v, x, y, z	the same as a, b, c, d, e, plus dots 3 and 6
w	2-4-5-6 — the exception. French braille had no w, so it was added later and does not fit the pattern. Learners always ask.

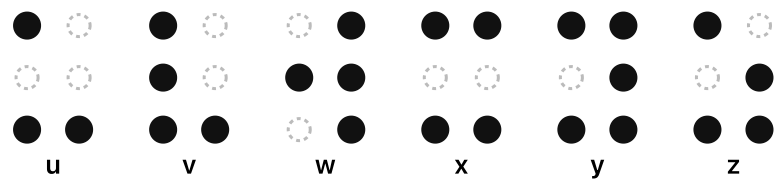
a – j (top four dots only)



k – t (a–j plus dot 3)



u – z (plus dots 3 and 6 — except w)



Two more for day one.

Capital sign — dot 6, placed before the letter.

Number sign — dots 3-4-5-6, after which **a–j** mean **1–0**.

capital

number

Confirm with the partner school which braille code they teach. This is standard English braille, but the school’s own practice is what the learner is examined on — they are the authority, not this sheet.

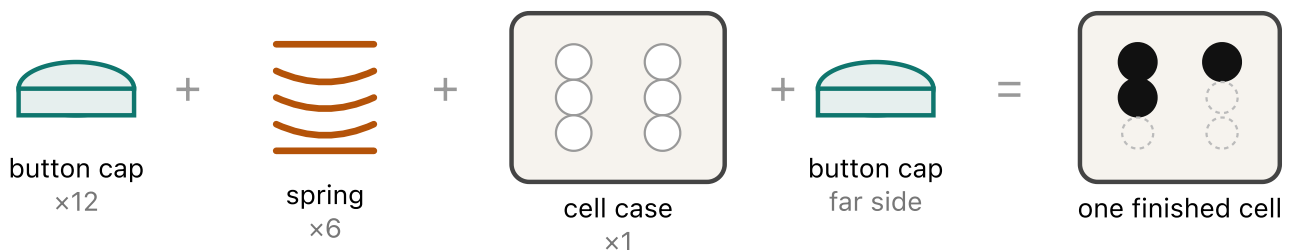
Printing

Part	Quantity per cell
Cell case	1
Spring	6 — one per dot
Button cap	12 — two per spring, one each side
Total	19 pieces per cell

The spring is the part that decides whether this works. Print it with **enough perimeter lines that the flexible arms come out solid.** If infill ends up inside the arms they will be weak and the button will not hold its position. This is the most common way the print fails.

- **No supports** are needed for any part.
- **Tested in PLA and PETG.** PETG is the better choice for something handled daily by a class; PLA is fine for a first trial.
- **Test print one spring and one button cap first** and check they seat properly in the cell case — before committing to the other seventeen pieces, let alone thirty sets.

Assembly



Per dot, **six times**:

1. **Snap a button cap onto one side of the spring.**
2. **Insert the spring into the cell case.** It only fits one way round.
3. **Press the button all the way in**, so the spring pops through the case and can be reached from the other side.
4. **Snap the second button cap on** that far side.

Both faces then have buttons, and pressing from either side toggles that dot.

Using it with a class

- **Set, feel, clear, repeat.** The rhythm matters more than the content. A learner who sets *a*, feels it, clears it and sets *b* has done more than one who has had ten letters read at them.
- **Teach the numbering before any letters.** A learner who knows the six positions can be taught a new letter over the phone.
- **Use the pattern openly.** Ten letters, then “add dot 3”, then “add dots 3 and 6”.
- **Sighted learners work with eyes shut, or the cell under the desk.** Otherwise they read the dots visually, learn nothing about touch, and leave believing braille is easier than it is.
- **Give a blind learner the cell first and longest.** For them this is the lesson itself, not an illustration of it.
- **Record it in the dashboard.** Session adaptations already include braille and tactile materials. Recording it is what turns “we make tactile materials” into a number we can report.

Related models by the same designer

Same author, so the same attribution applies. The millimetre figures are **dot spacing**.

Model	Where
Braille Training Cell — 6MM	Printables 807178
Braille Training Cell — 4.5MM	Thingiverse 6538334
Braille Sheet — 4.5MM	Printables 1231140

Standard braille is smaller than either figure. Oversized cells are kinder to a beginner’s fingers and to a sighted teacher, but a learner has to reach standard spacing eventually. Decide deliberately whether we are teaching braille or teaching *about* braille — and ask the partner school which they want.

Before the first visit

☐	Print one spring + one button cap and check the fit. Do not skip.
☐	Print and assemble one complete cell. Time yourself — 19 parts.
☐	Check the buttons still hold position after ~50 presses.
☐	Decide PLA or PETG for the class set.
☐	Agree dot spacing with the school.
☐	Work out print time and filament cost per cell, then per class.
☐	Note the attribution to 3D Printy wherever the object appears.

Deliberately left blank. Dot spacing, overall dimensions, print time and filament cost per cell are not stated here. They should be **measured from our own first print** rather than asserted from a general claim — and once measured, they belong on this sheet. The diagrams above are drawn for this guide, not taken from the model page; photographs of our own printed cells should replace nothing here but should be added alongside.